Basics of Chemotherapy
Anti-malarias & HIV Treatment

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Basics of Chemotherapy

3 Parts

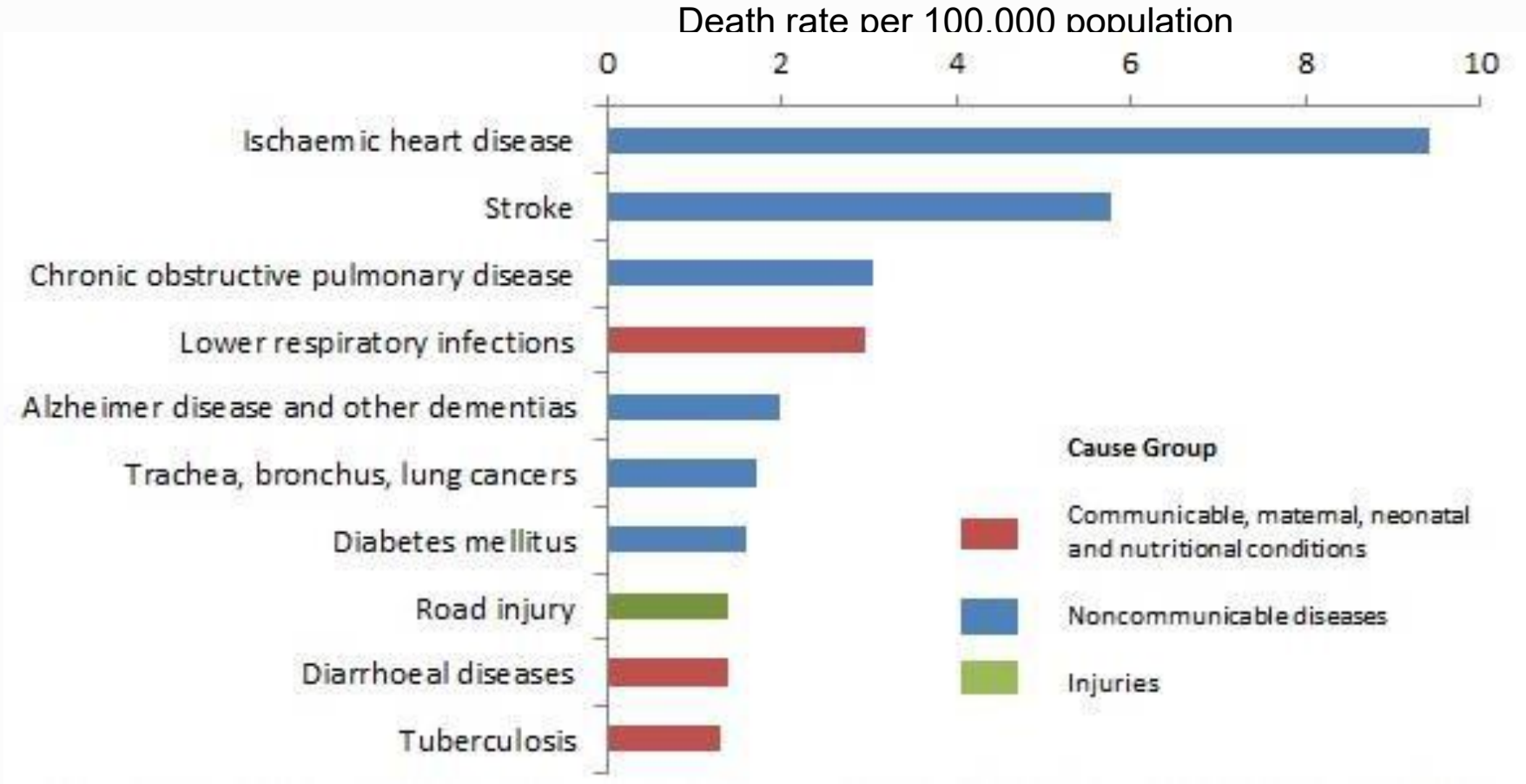
1) To understand the basic concept underlying the use of chemotherapy to treat diseases.

2) To understand chemotherapeutic treatment of Malaria

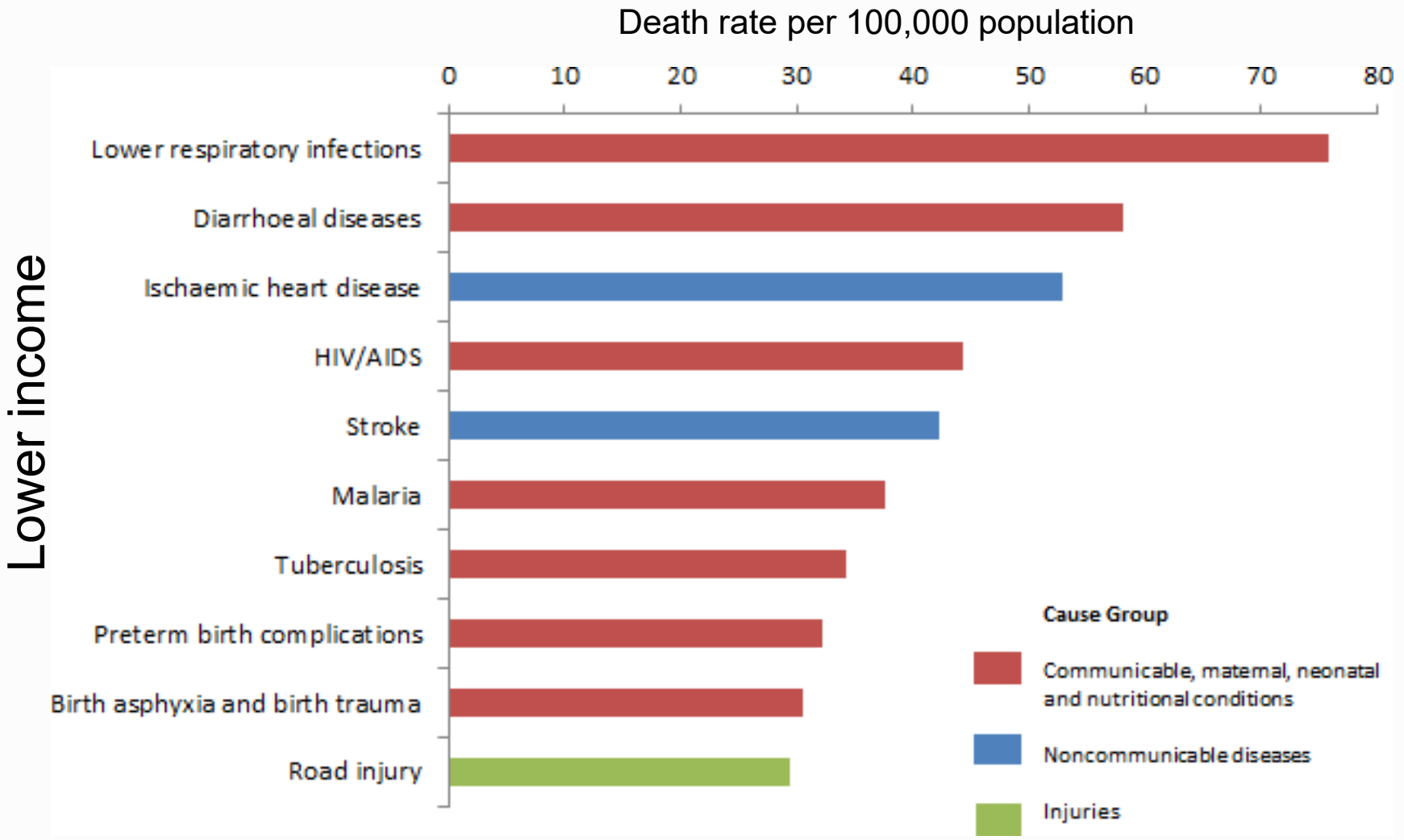
3) To understand chemotherapeutic treatment of HIV

🟩 Pharmacology. Rang, Dale & Ritter 5th Edition Chapters 44, 46 & 48.

Differences in cause of death around the world 2016



Differences in cause of death around the world 2016



History of Chemotherapy



Paul Ehrlich 1854-1915
Nobel Laureate 1908

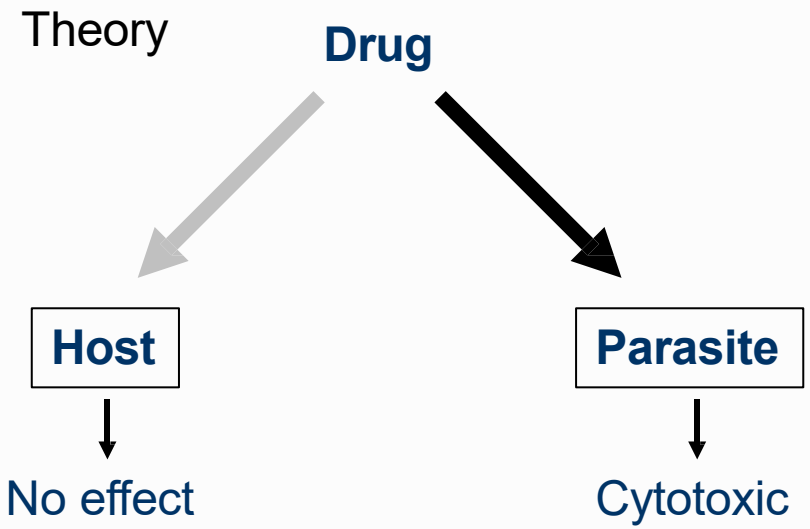
- If we picture an organism as infected by a certain species of bacterium, it will . . . be easy to effect a cure if substances have been discovered which have a specific affinity for these bacteria and act…on these alone. . . while they possess no affinity for the normal constituents of the body. . . such substances would then be . . . magic bullets

Gertrude Belle Elion & George Hitching Continued this work and together with James Black was awarded 1988 Noble prize

Theory of Chemotherapy. The Idea.

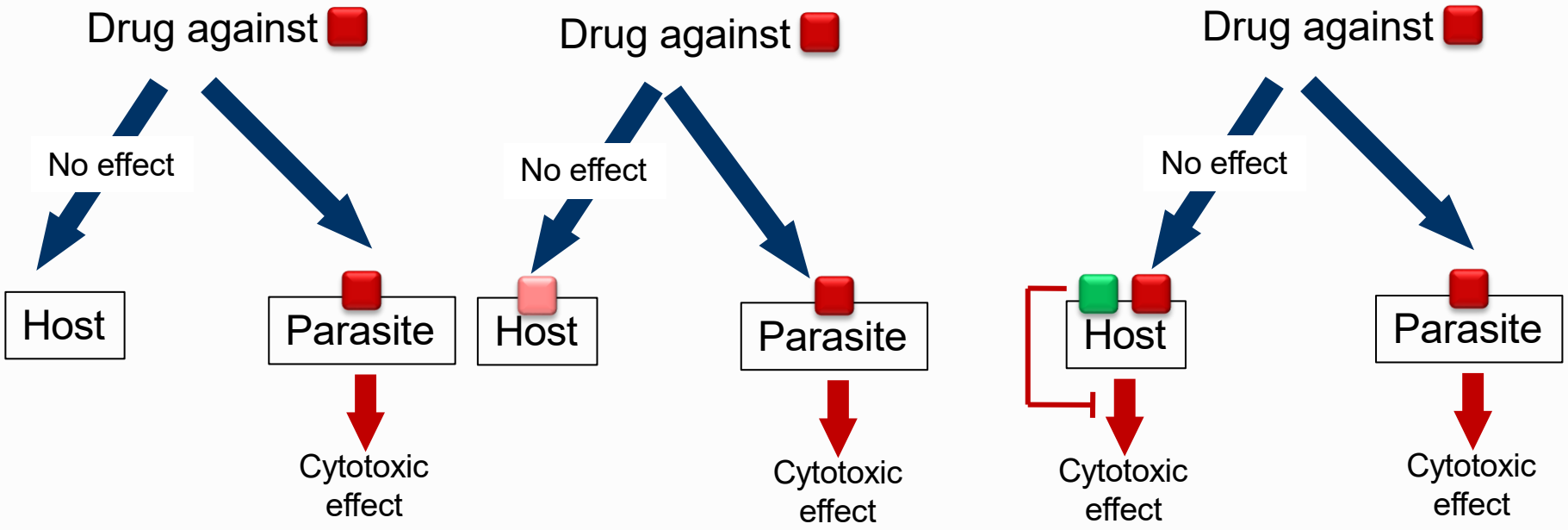
For the case of Chemotherapy bacteria, fungi, protozoa, helminths, viruses and cancer cell are considered parasites.

Find Qualitative (preferable) or Quantitative Biochemical difference between Host and Parasite which when exploited by a selective drug results in a cytotoxic effect to the parasite but not host



With the on-set of genomics/proteomics and systems Biology the theory of Chemotherapy could be expanded to systemic chronic pathologies

Variations on a theme



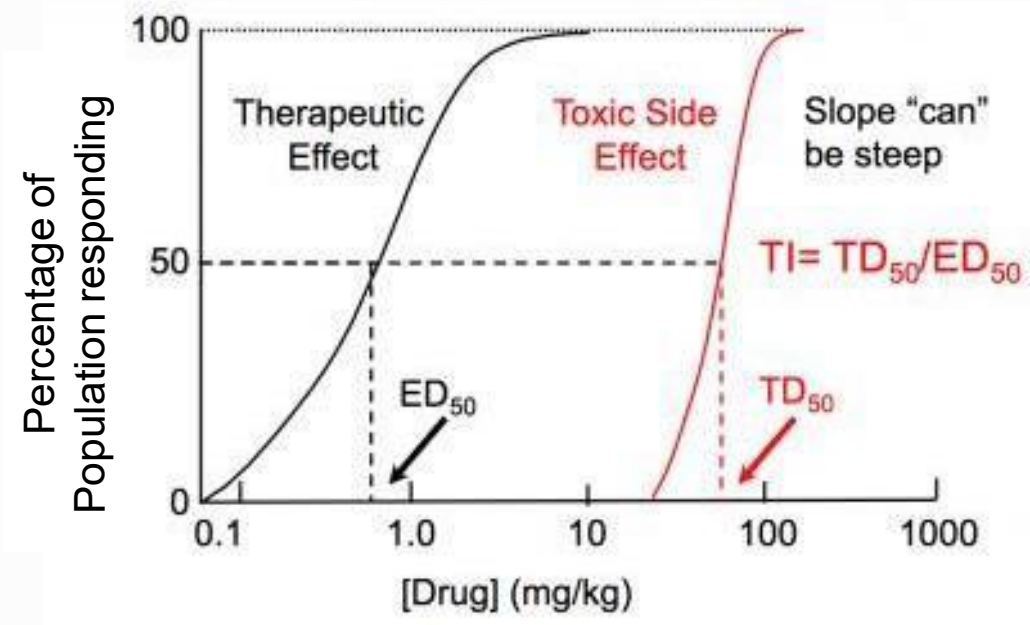
Therapeutic index (TI)

A measure of drug safely based on populations.

ED_{50}. (Median effective dose). Is the dose that produces the therapeutic effect in in 50% of the population.

TD_{50}. (Median toxic dose). Is the dose that produces the toxic effect in 50% of the population.

LD_{50}. (Median lethal dose). Is the dose that produces the Lethal effect in 50% of the population.



Humans

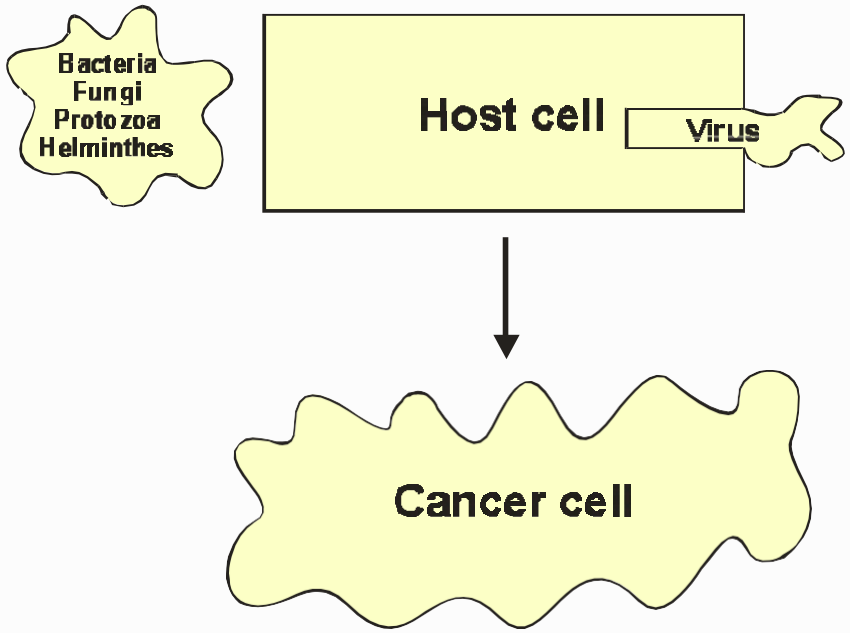
$$TI = \frac{TD_{50}}{ED_{50}}$$

Animals

$$TI = \frac{LD_{50}}{ED_{50}}$$

Theory of Chemotherapy. The problems

- Bacteria/Protozoa have many differences, Viruses have fewer (hijacking some of the hosts biochemistry) Cancer is essentially self (rely on quantitative differences



- Development of resistance (adaptation/ mutation)

Theory of Chemotherapy. The Targets.

- **Class I Biochemical reactions (energy production)**

  Gycolysis, Catabolism of polysaccharides and lipids, pentose phosphate pathway, pyruvate oxidation, citric acid cycle, fermentation and respiration

- **Class II Biochemical reactions (Small molecule synthesis)**

  Biosynthesis of amino acids, Lipids, Coenzymes (tetrahydrofolate), Purines & Pyrimidines, heme catabolism/metabolism

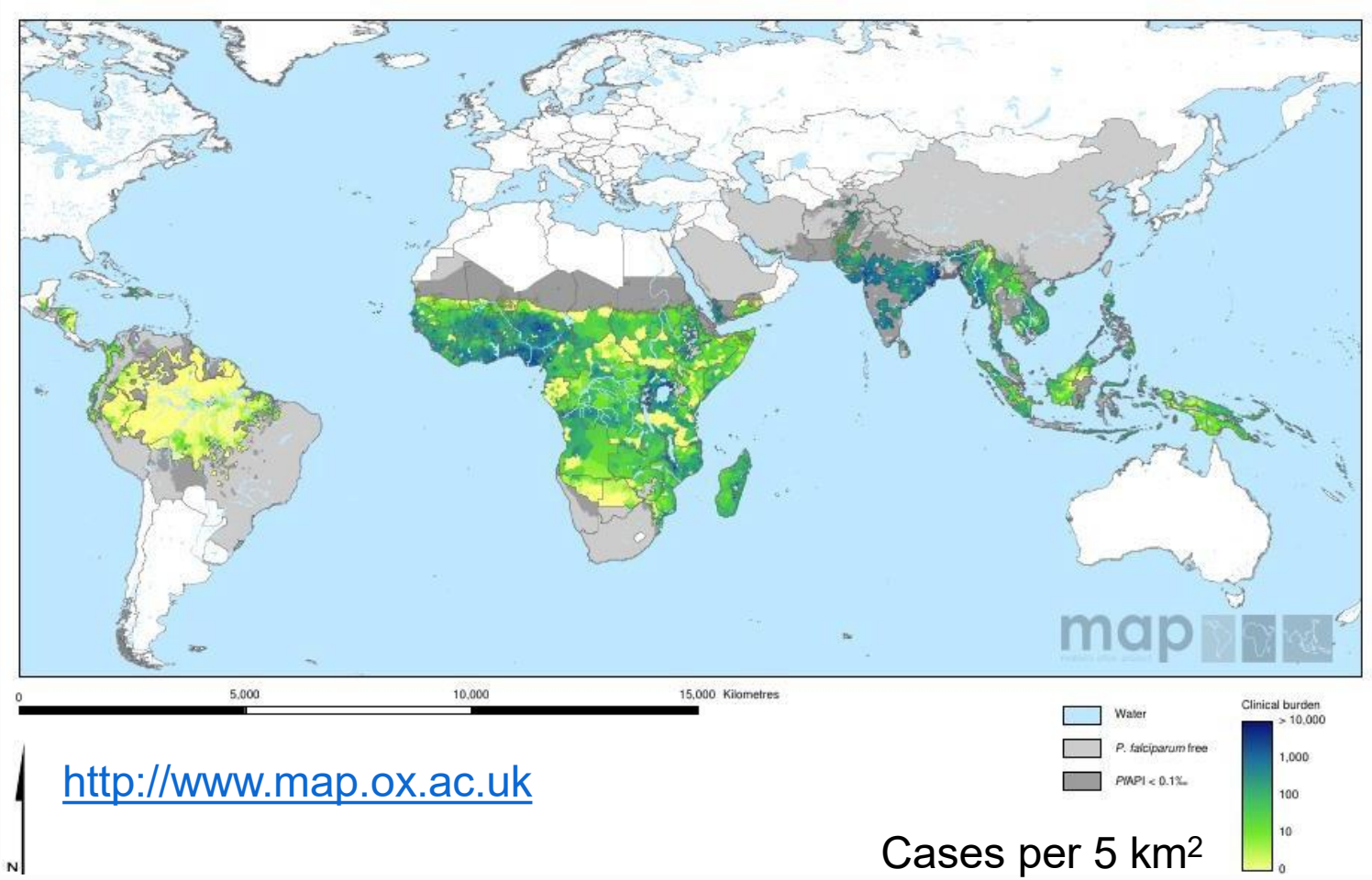
- **Class III Biochemical reactions (Macromolecule production, polymers)**

  Biosynthesis of DNA, RNA, proteins, Lipoproteins, peptidoglycans protein degradation, Cytoskeleton assembly, Glycogen synthesis,

Malaria

- Protozoa: unicellular eukaryotics

Plasmodium falciparum: Most serve form no exoerythrocyte stage
Plasmodium vivax: less serve than *P. falciparum* does have exoerythrocyte stage
Plasmodium ovale: Rare does have exoerythrocyte stage
Plasmodium malariae: Widespread but no exoerythrocyte phase

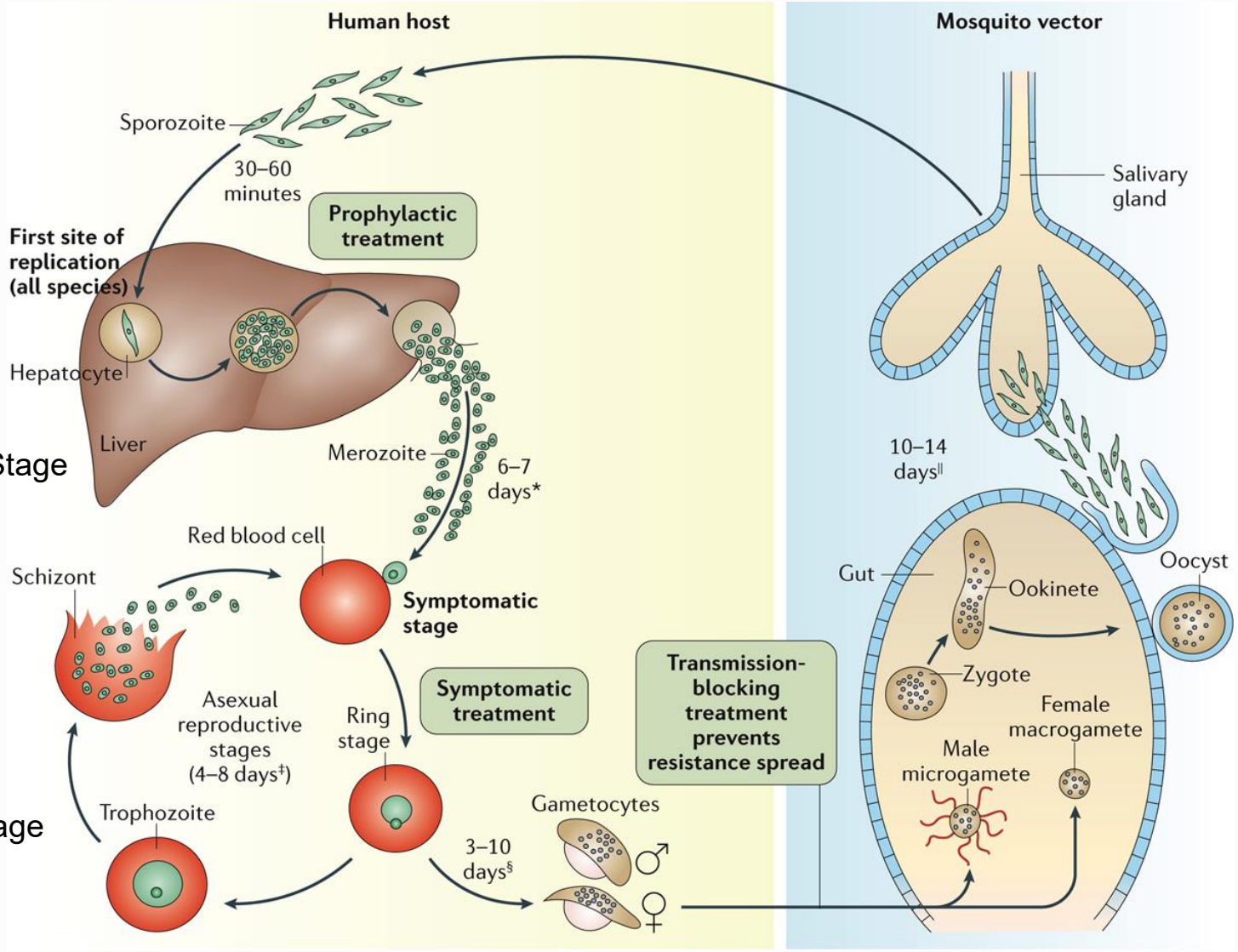


http://www.map.ox.ac.uk

Cases per 5 km^2

Malaria

- Life cycle



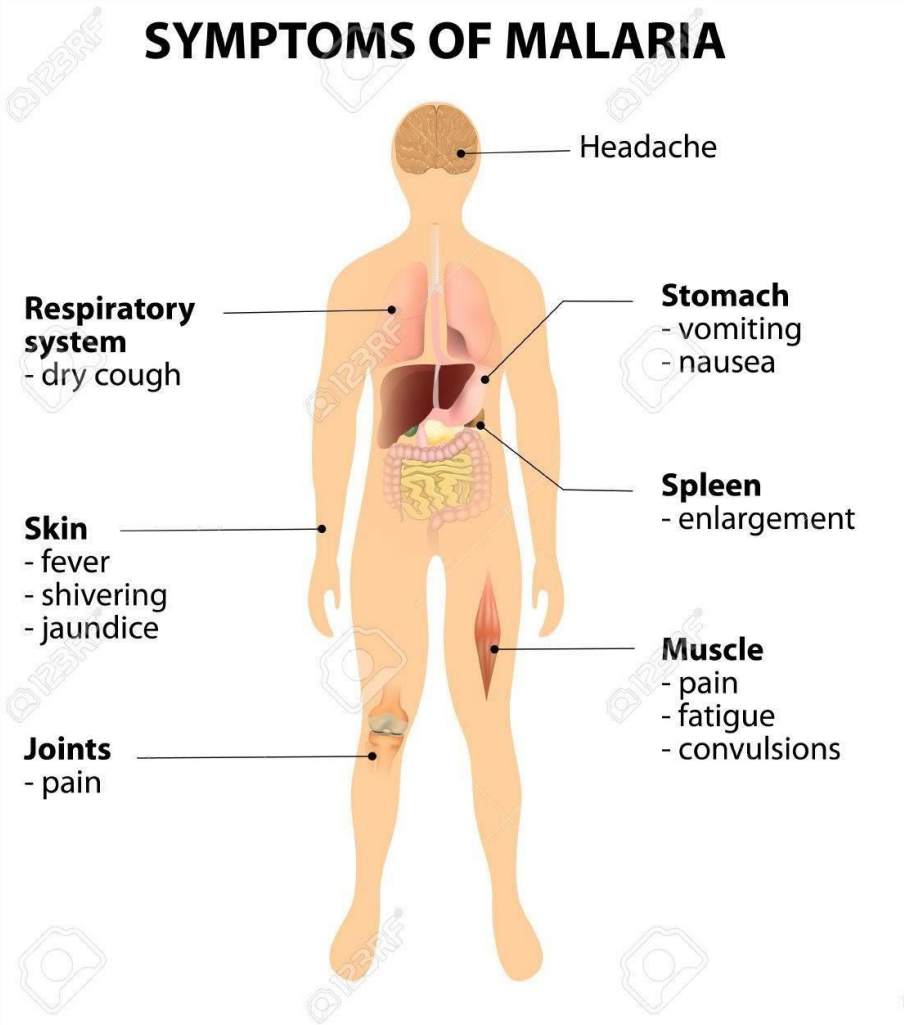
Malaria

- Disease

Uncomplicated Malaria
Severe Malaria
Malaria Relapses

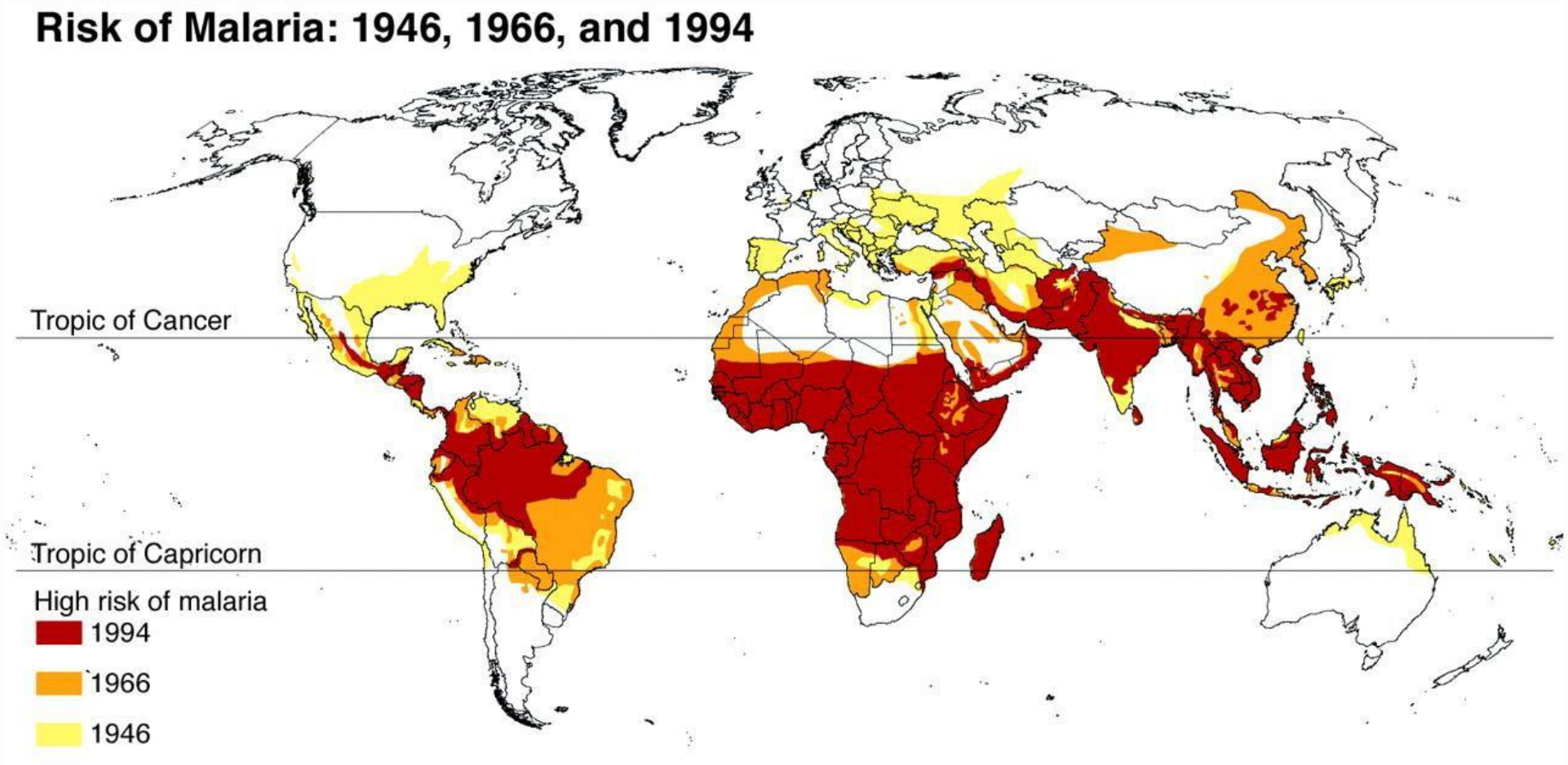
- Malarial species, strains
- Drug metabolism
- Side effects
- Host's immunity

- Protect (prophylactic)
- Cure established infection
- Prevent malarial transmission

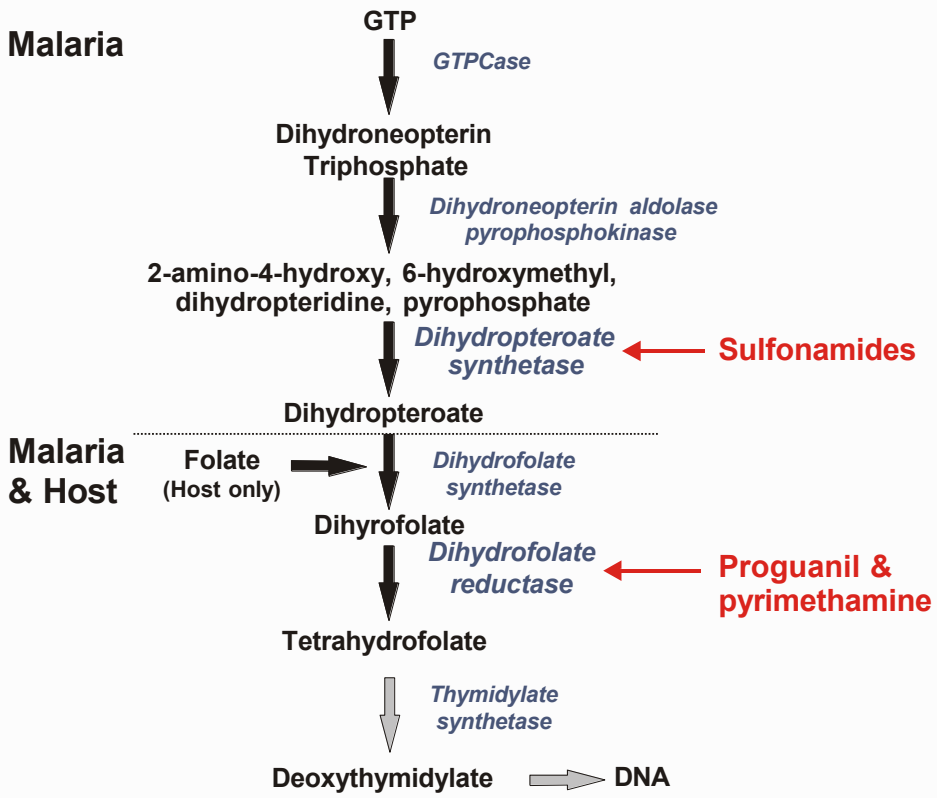


Prevention of Malaria.

Irrigation, pesticides & education



Malaria. Anti-malarial drugs. The Folate pathway

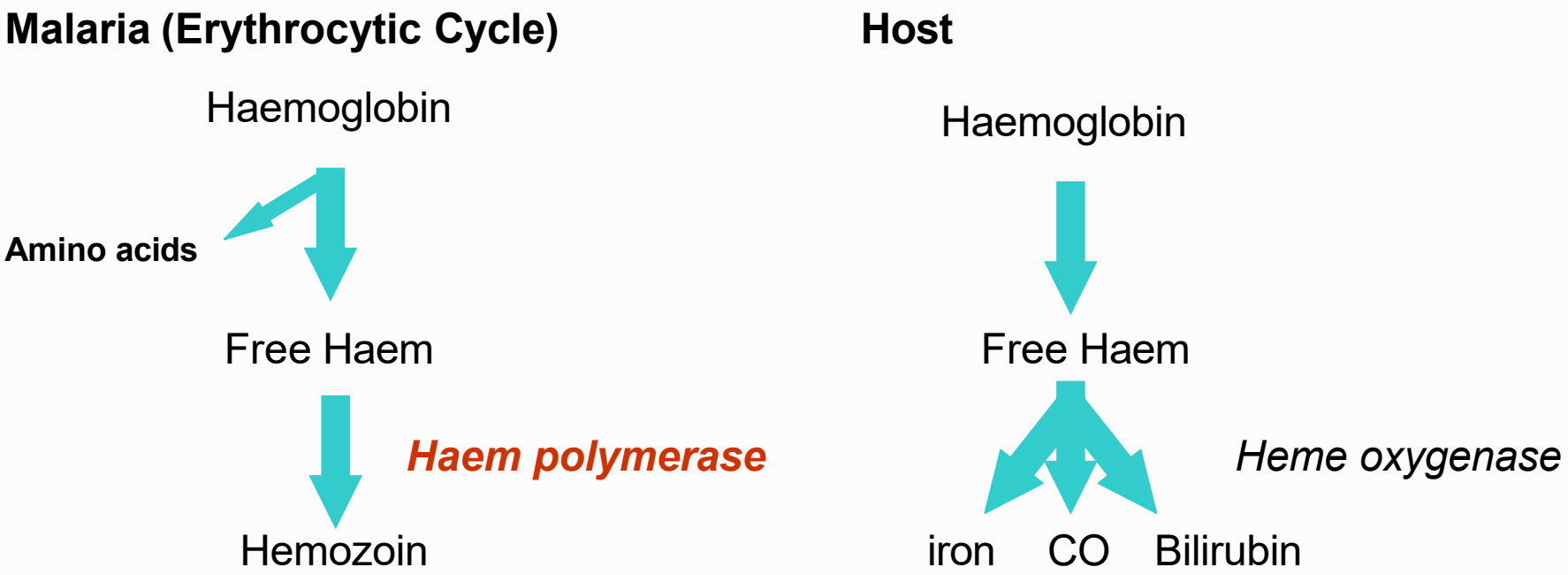


	C_{50} (μmol/l) for FH_2 reductase		
	Human	Protozoal	Bacterial
Trimethoprim	260	0.07	0.005
Pyrimethamine	0.7	0.0005	2.5
Methotrexate	0.001	0.1	inactive

- Sulphonamides & Sulphones. Rely on the fact that malaria synthesis folate and host does not
- Proguanil & pyrimethamine. Rely on the different sensitivity of malaria and host dihydrofolate reductases to antagonists

Malaria. Anti-malarial drugs. Heme catabolism

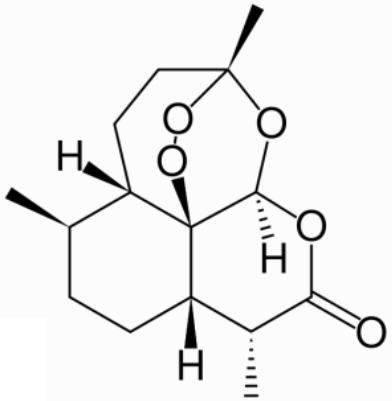
- Chloroquine & quinine target another qualitative difference between malaria and host, that of heme catabolism

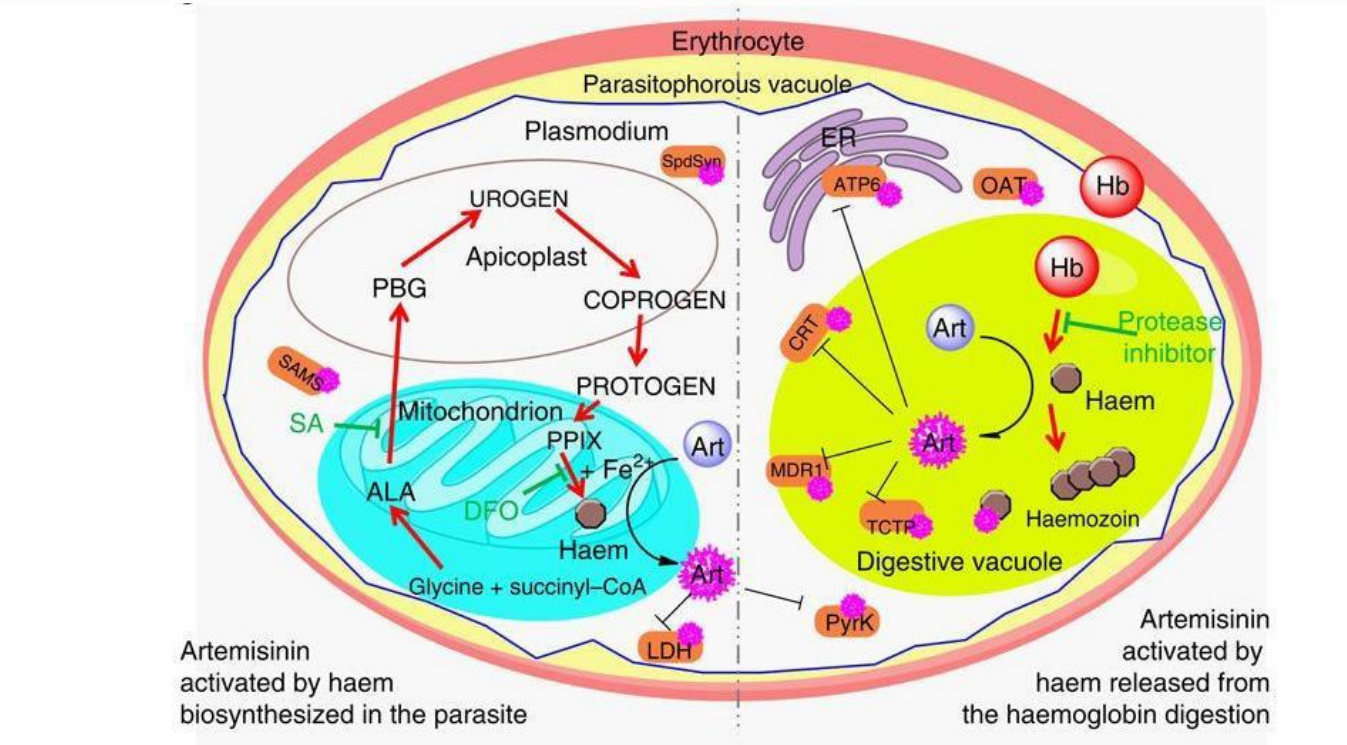


Mefloquine mechanism not known

Artemisinin (Artesunate)

- Isolated from sweet wormwood (*Artemisia annua*)
- Used in combination therapy e.g. Artesunate–mefloquine
- Mode of action is complex and relies on Artemisinin activation/metabolism



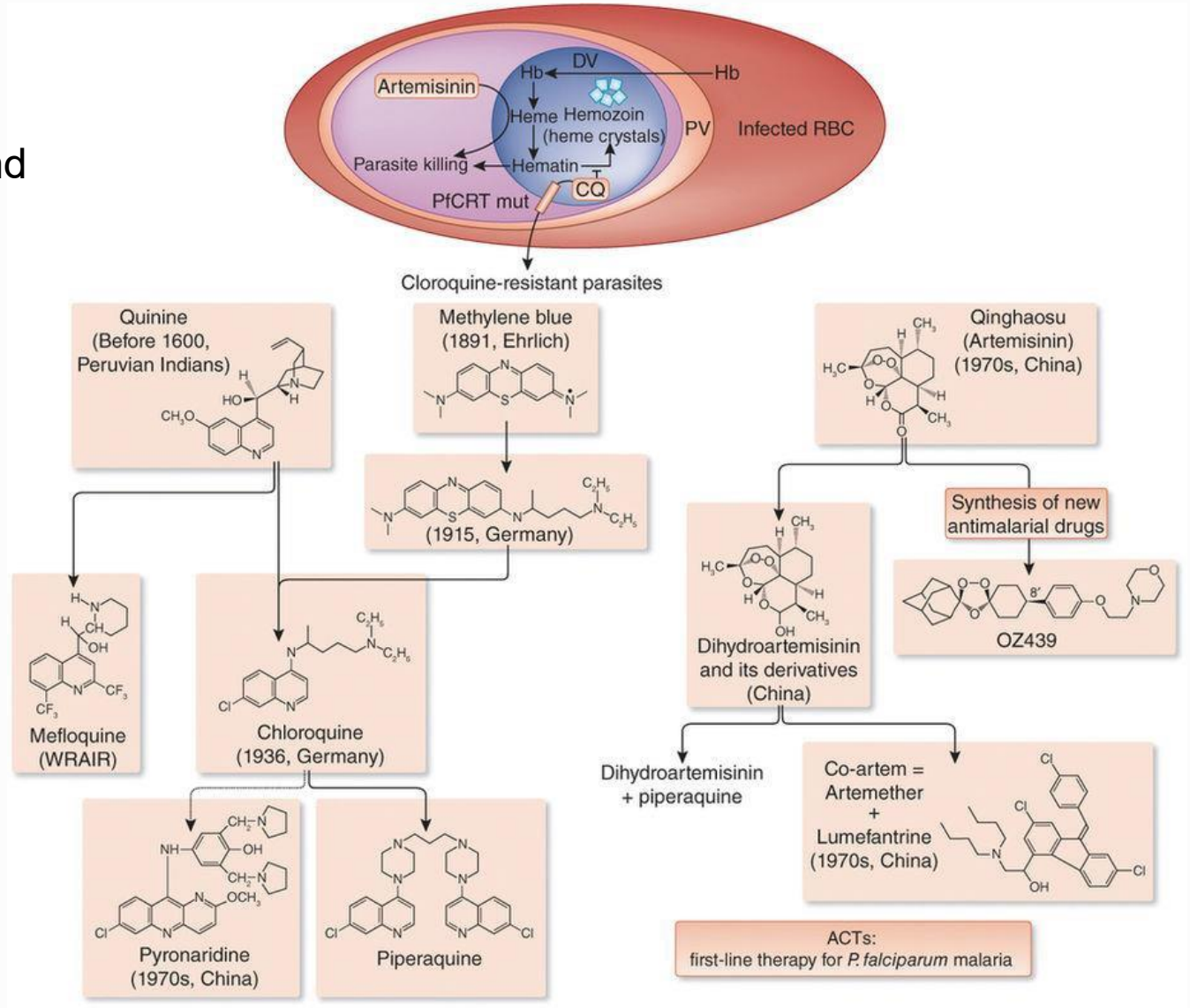


Noble prize (co-recipint) in Medicine & Physiology (2015) went to Tu Youyou of the China Academy of Traditional Chinese Medicine for her work on Artesunate.

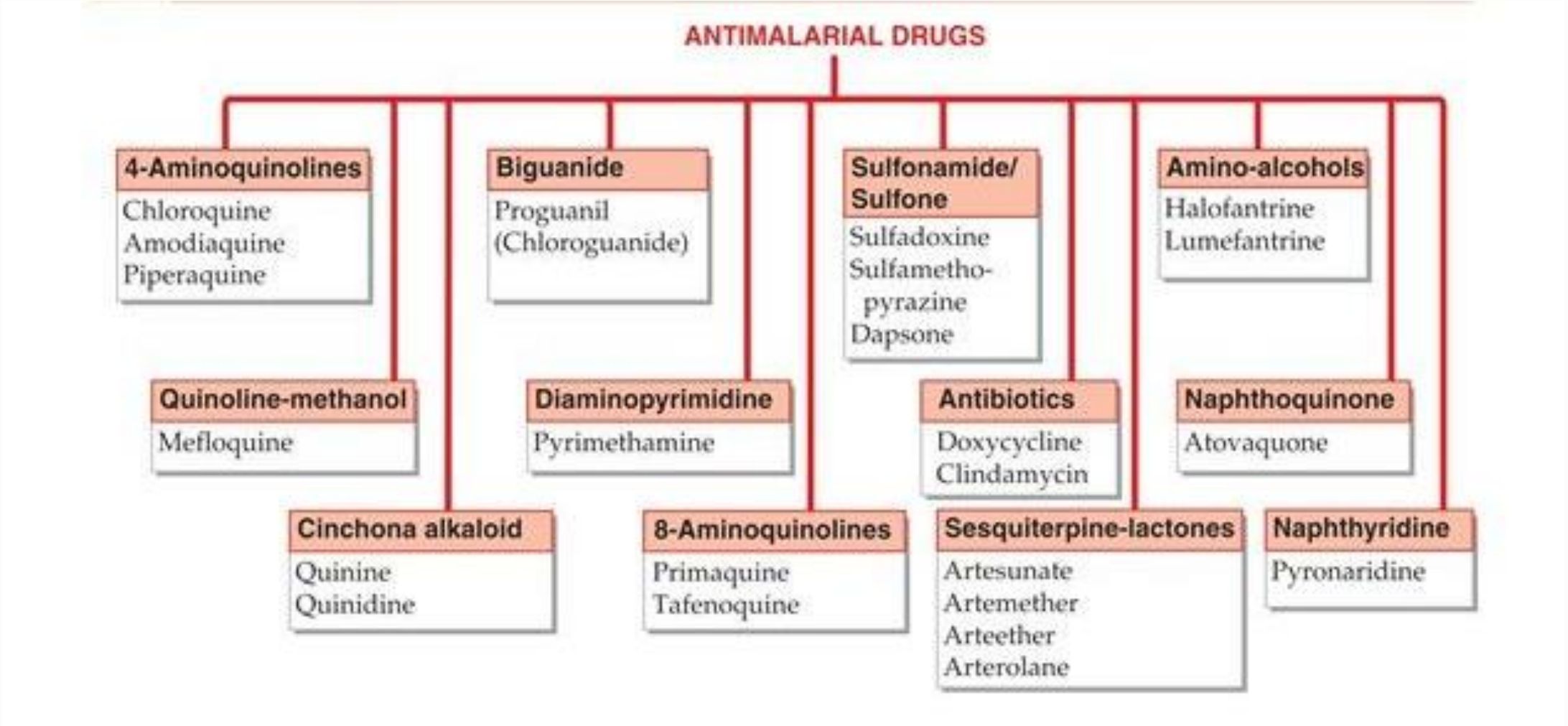
Artemisinin (Artesunate)

Quinine and artemisinin discoveries led to the development of many synthetic antimalarial drugs

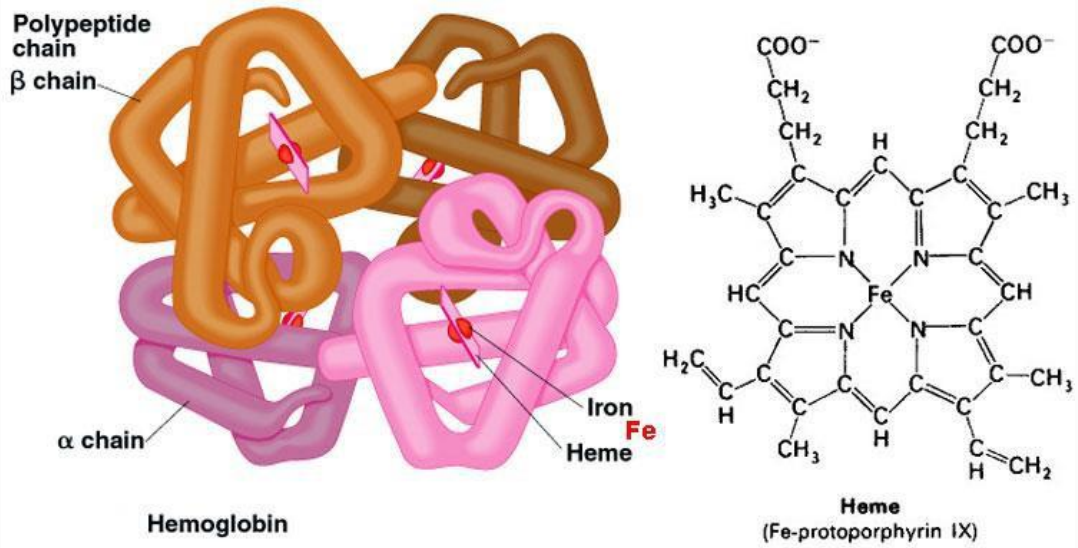
- Natural compound
- Semi-synthetic compound
- Synthetic compound



Drugs against Malaria



Pharmacogenomics & Malaria



Natural resistance

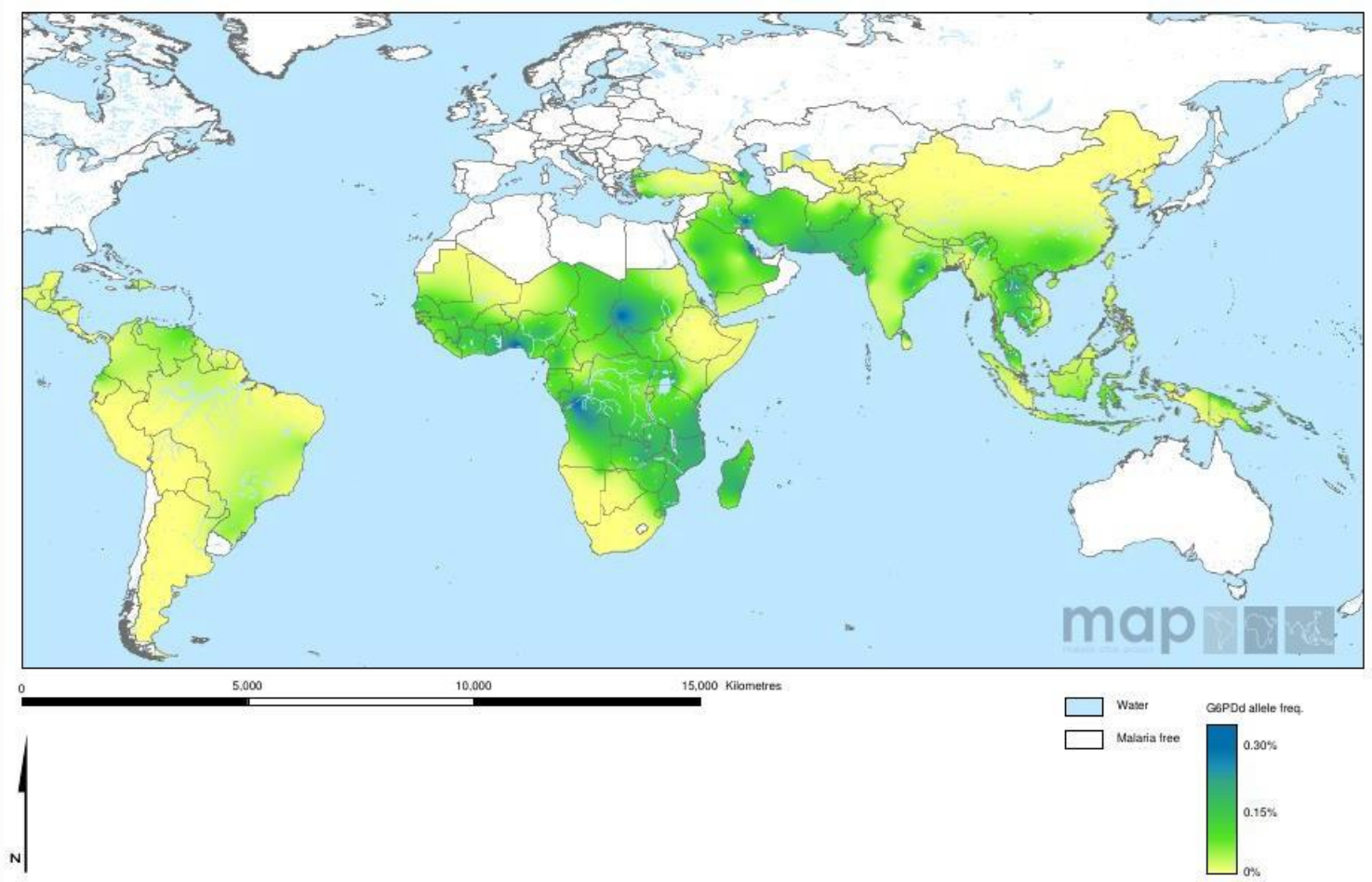
Sickle cell anaemia

Thalassemia

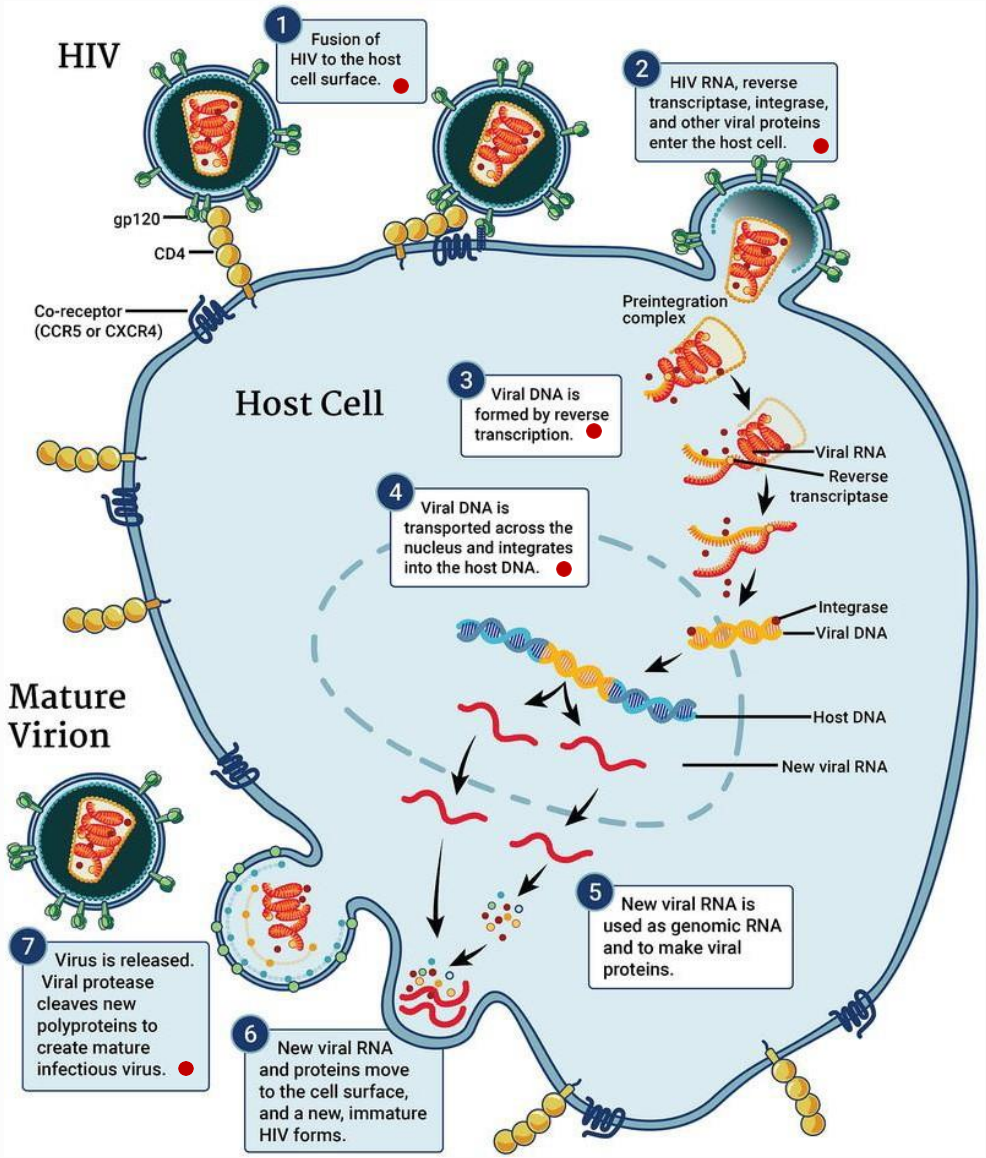
Glucose 6 phosphate dehydrogenase deficiency

Anti-malaria drug cause anaemia

Median predicted allele frequency for G6PD deficiency map in 2010 globally



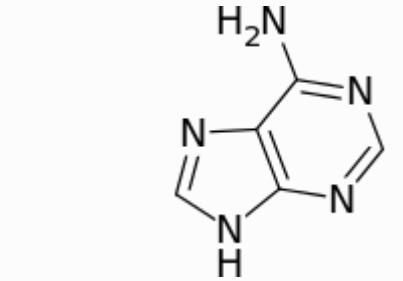
HIV infection



Retrovirus
Positive-strand RNA
15 genes

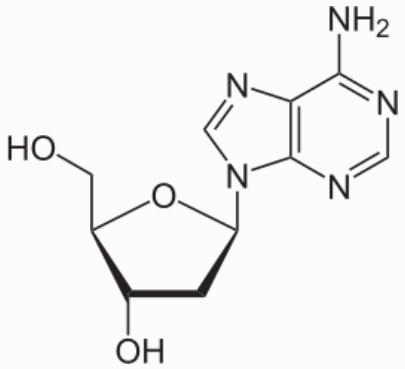
- Intracellular viral replication
- Infection is not detected until late in the viruses time course
- Rapid viral evolution

Nucleotide structure



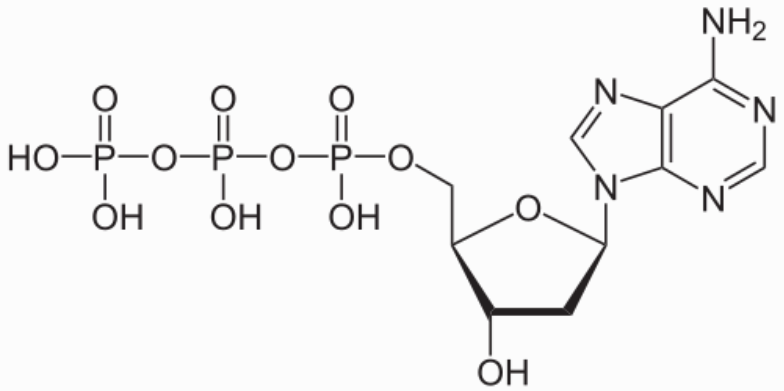
Adenine

Pyrimidine	Purine
Uracil	Adenine
Cytosine	Guanine
Thymine	



Deoxyadenosine

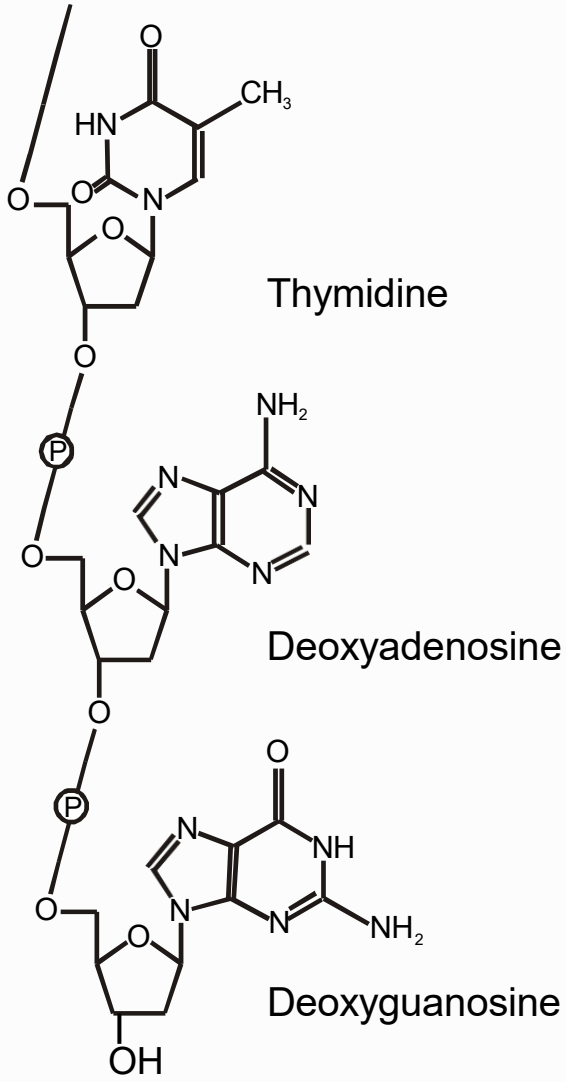
BASE + SUGAR = NUCLEOSIDE

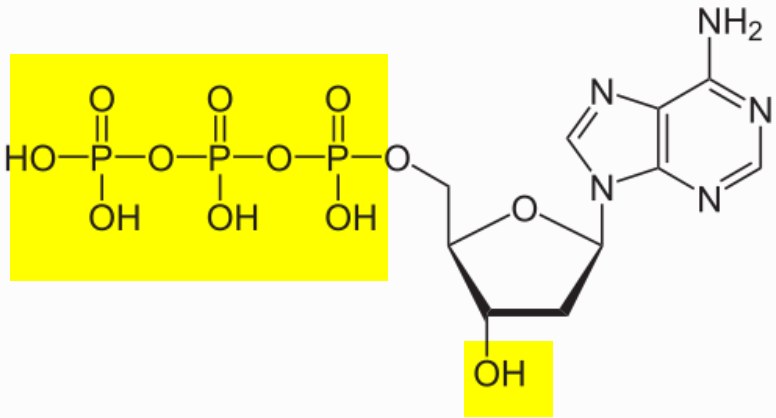


Deoxyadenosine triphosphate

BASE + SUGAR + PHOSPHATE = NUCLEOTIDE

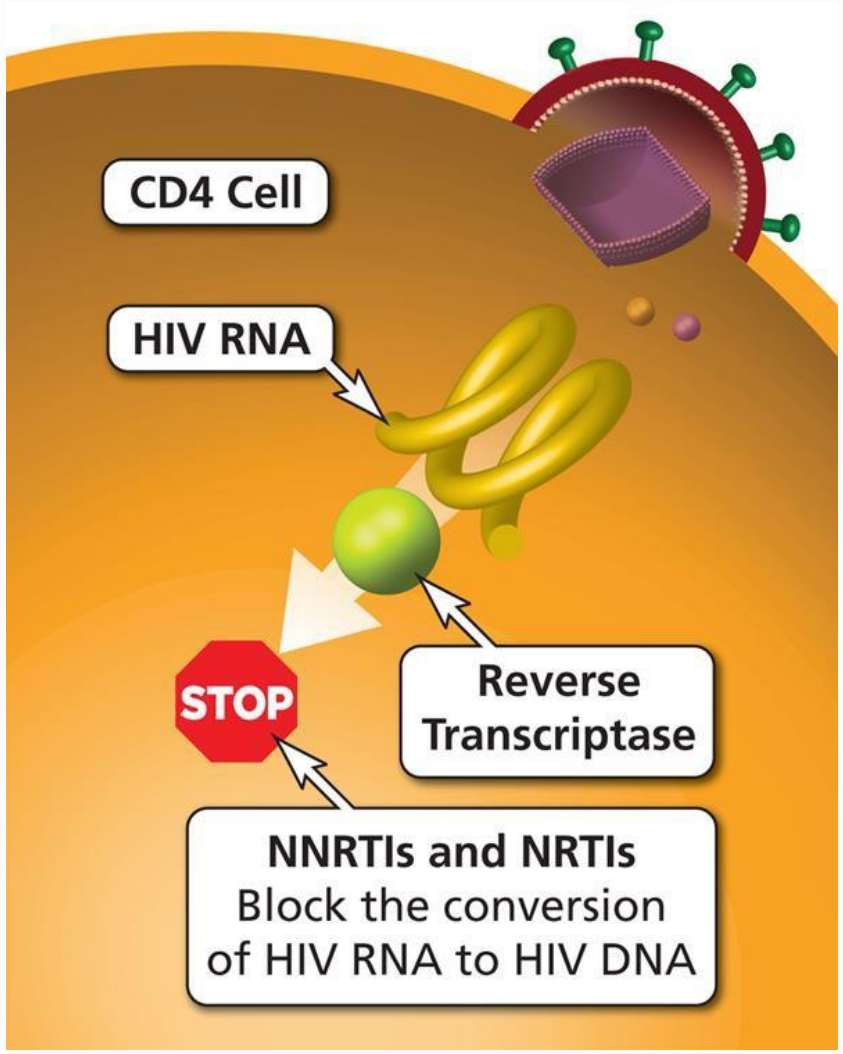
DNA backbone Structure



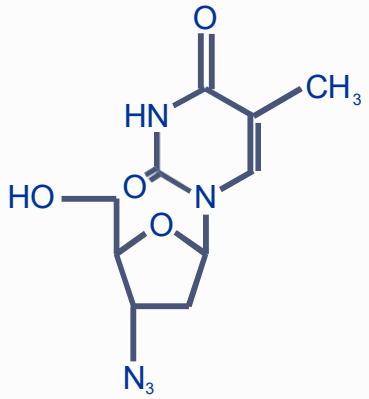


Reverse transcriptase (RT)

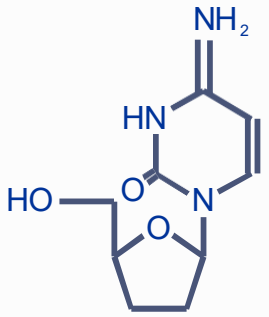
Two types of inhibitors. Nucleoside RT Inhibitors & Non-nucleoside RT Inhibitors



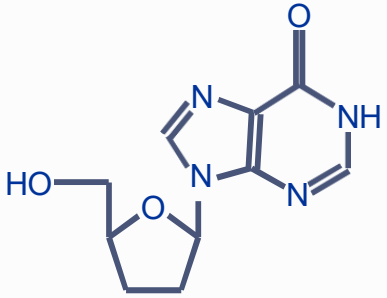
Nucleoside Reverse Transcriptase inhibitors



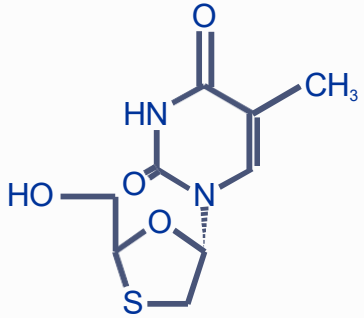
AZT
3’-azido-3’ deoxythymidine



DdC
2’, 3’-Dideoxycytidine

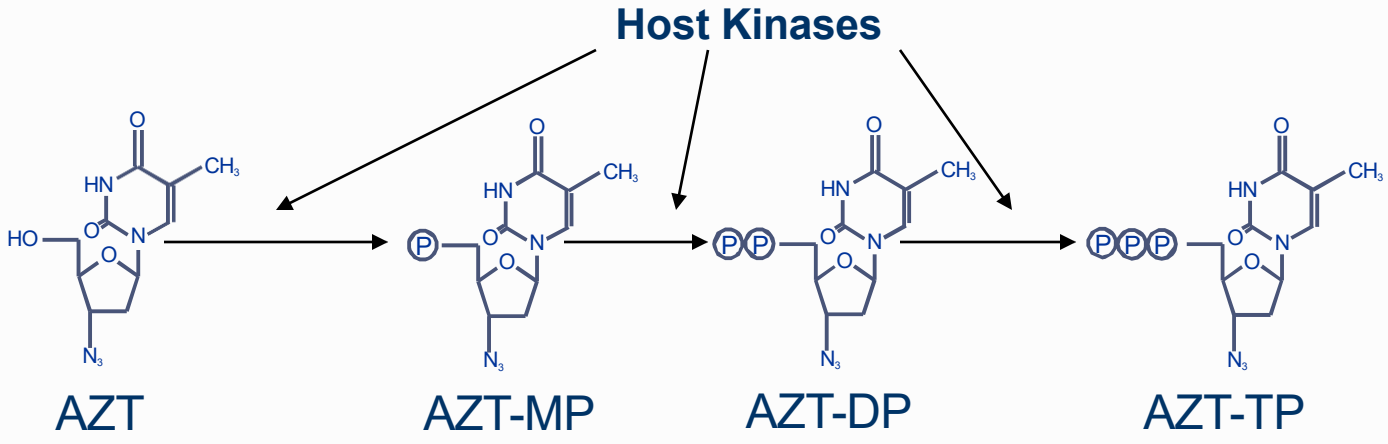


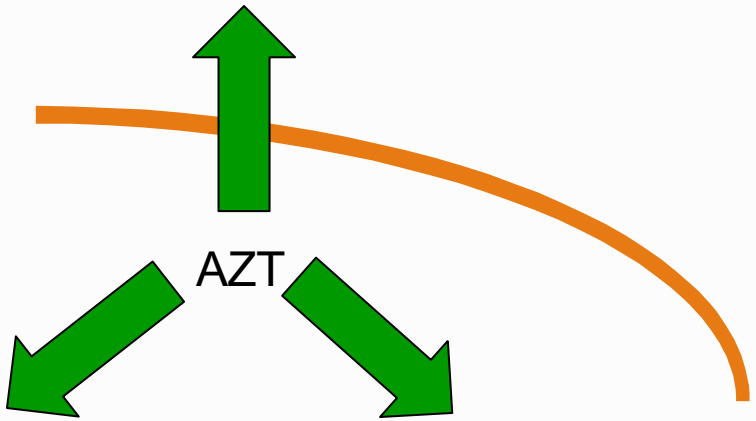
DdI
2’, 3’-Dideoxyinosine

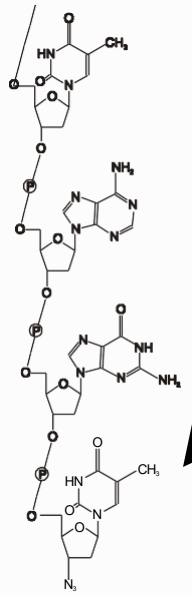


3TC
3’-thiabofuranosyl-ᛋL-cytosine

Nucleoside Reverse Transcriptase inhibitors
mechanisms of action in HIV





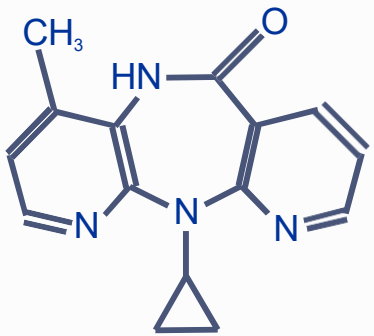


Viral Reverse Transcriptase

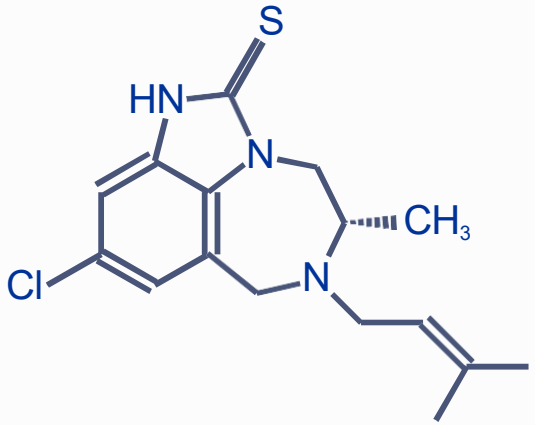
① Host DNA polymerase

Infected Cell

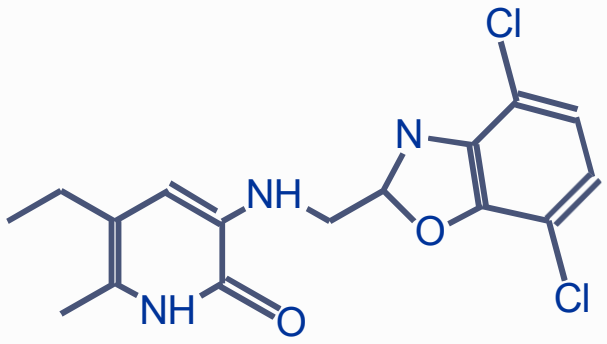
Non-nucleoside Reverse Transcriptase inhibitors (NNRTIs)



Nevirapine

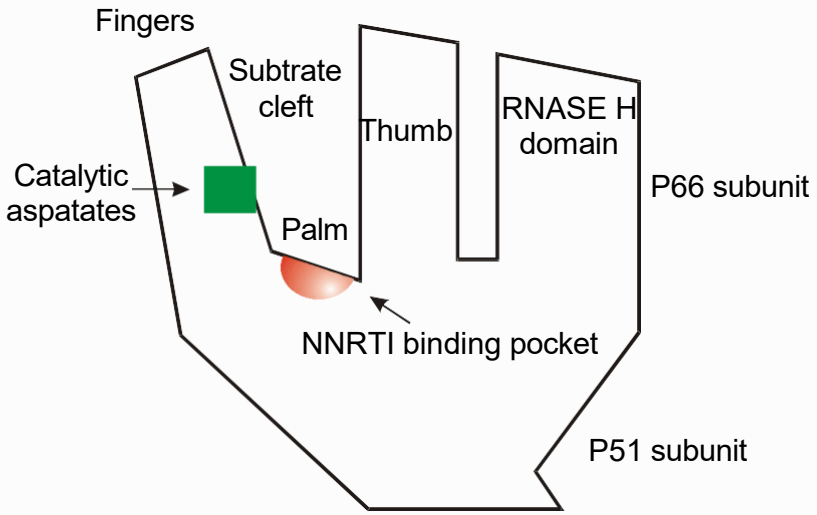


C1-TIBO

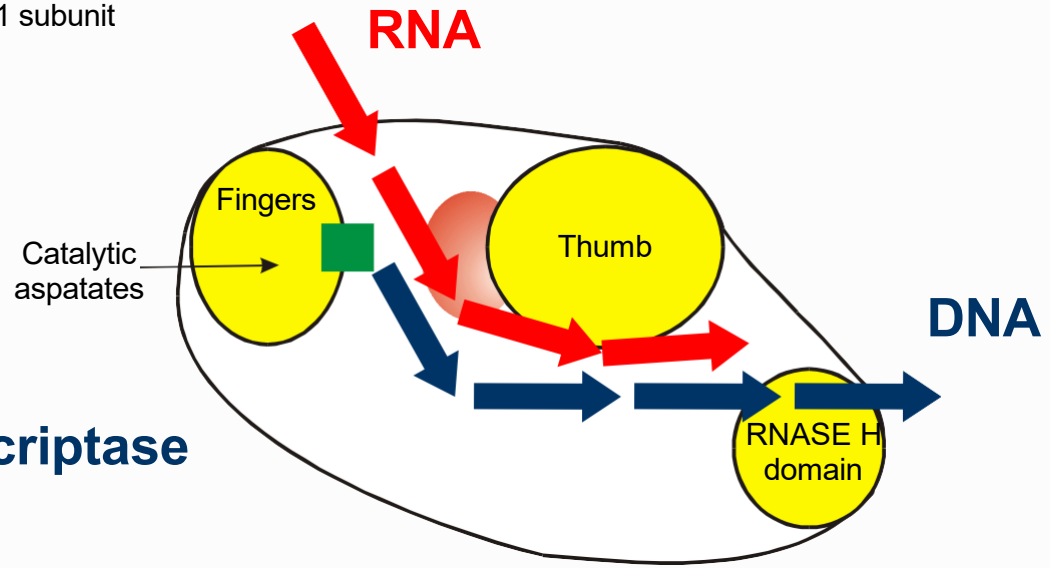


L697-661

Non-nucleoside Reverse Transcriptase inhibitors (NNRTIs)
mechanisms of action in HIV

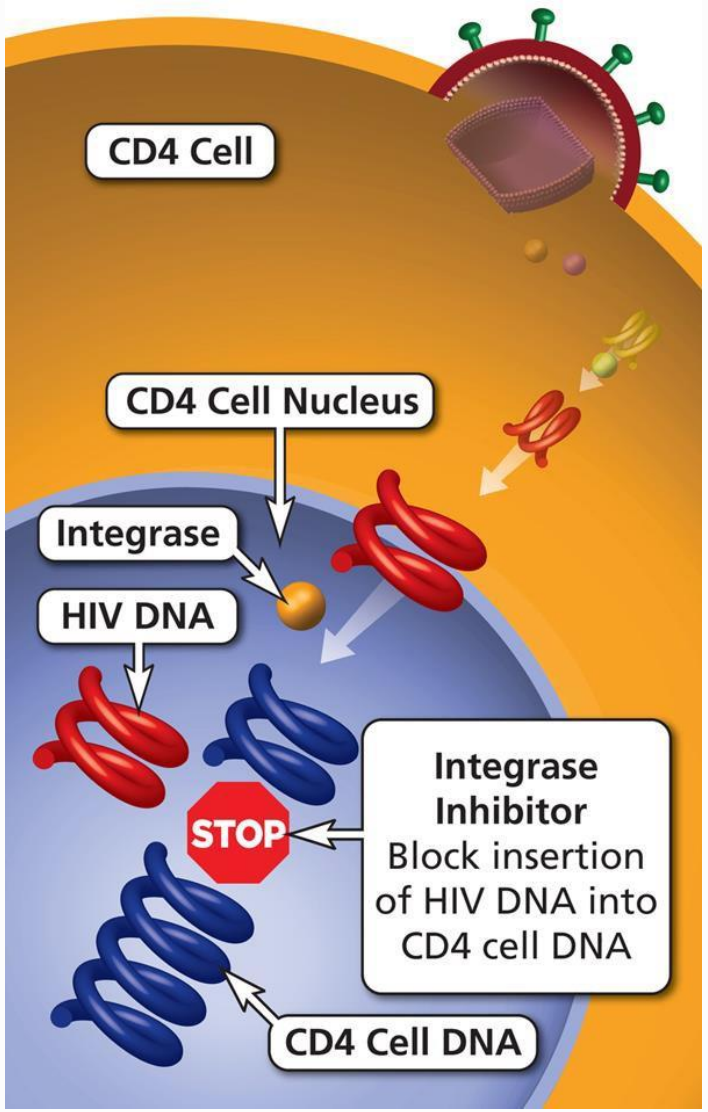


Side-view Reverse Transcriptase

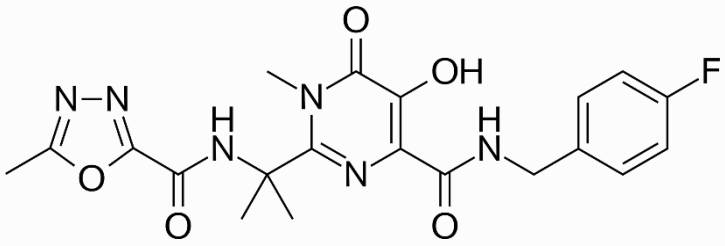


Top-view Reverse Transcriptase

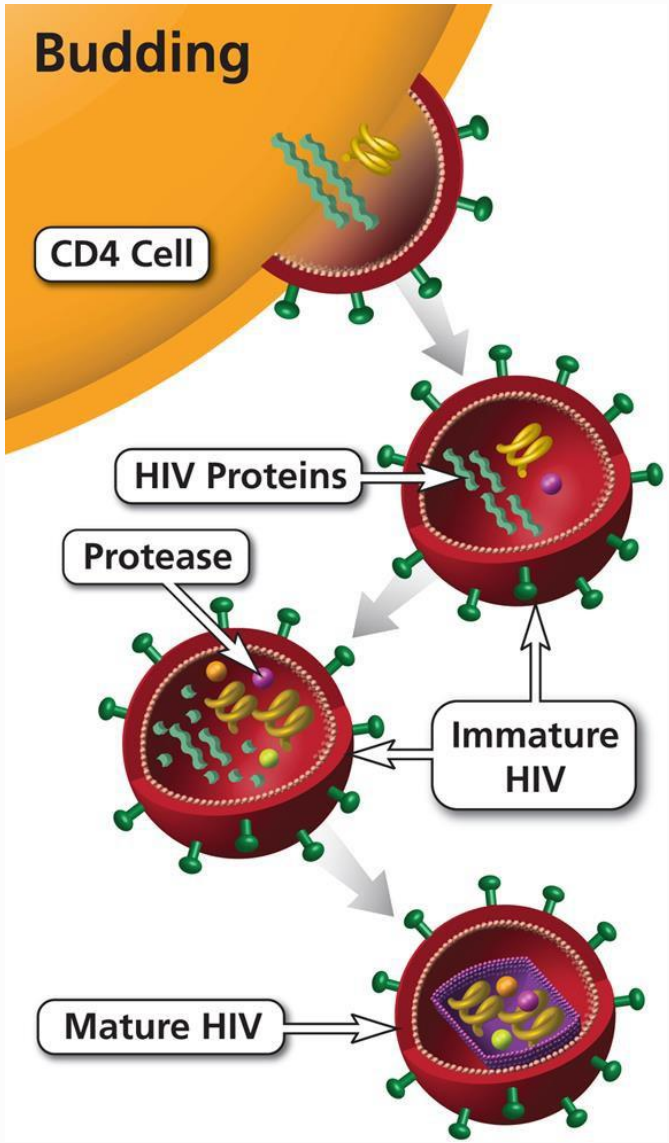
Integrase Inhibitors



Raltegravir



HIV Protease (aspartly protease) inhibitors

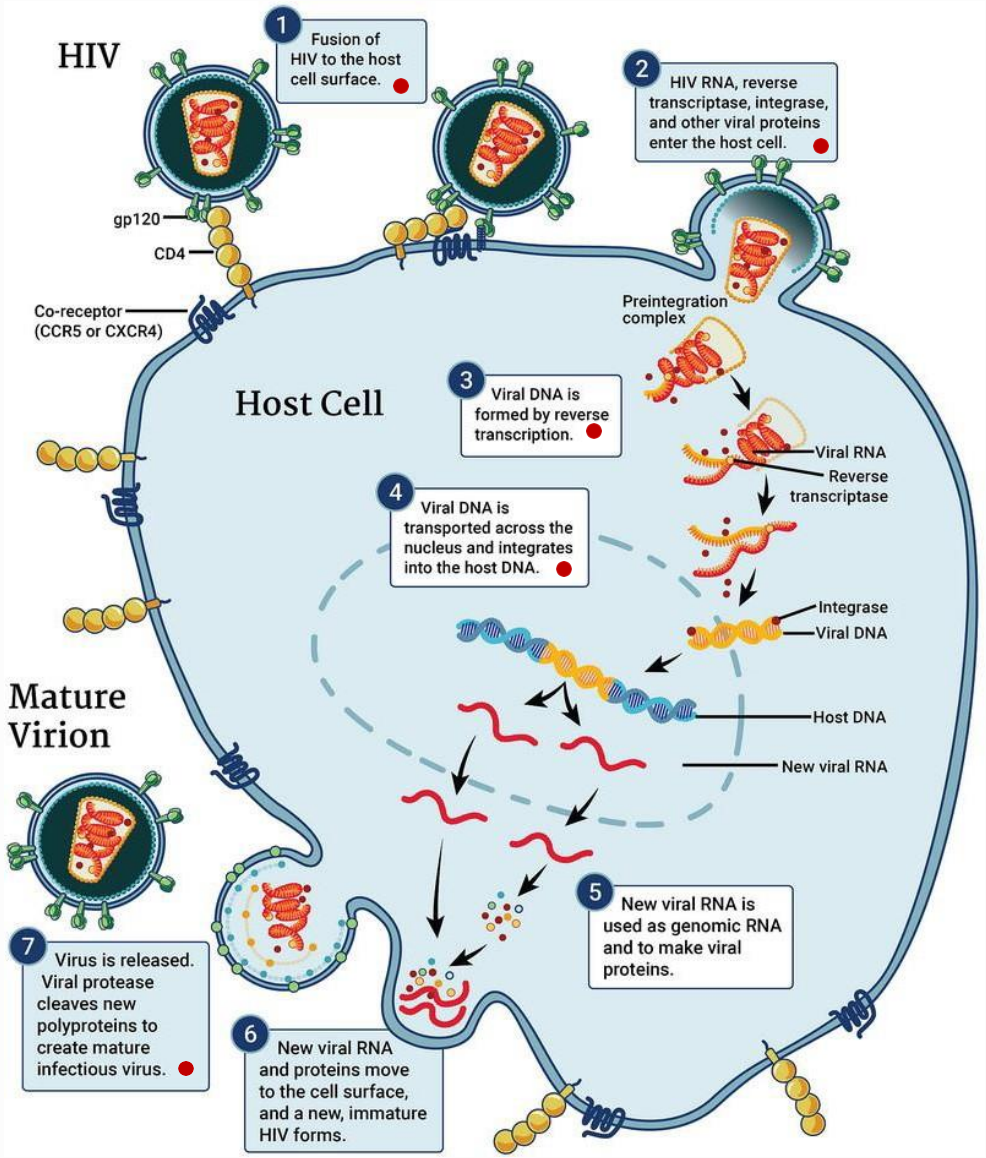


Viral Protease required for the splicing of viral proteins

GAG	POL	ENV

e.g. Indinavir

HIV infection



Retrovirus
Positive-strand RNA
15 genes

- Intracellular viral replication
- Infection is not detected until late in the virus's time course
- Rapid viral evolution

Next lecture we will look at

- Anti-bacterial agents
- Criteria for selecting anti-bacterials and resistance
- Theory of chemotherapeutic agents' combinations

Anti HIV therapies

NRTI

Abacavir, ABC (Ziagen®)
Didanosine, ddI (Videx®, Videx EC®)
Emtricitabine, FTC (Emtriva®)
Lamivudine, 3TC (Epivir®)
Stavudine, d4T (Zerit®)
Tenofovir alafenamide, TAF*
Tenofovir disoproxil fumarate, TDF (Viread®)
Zidovudine, AZT/ZDV (Retrovir®)

NNRTI

Delavirdine, DLV (Rescriptor®)
Efavirenz, EFV (Sustiva®)
Etravirine, ETR (Intelence®)
Nevirapine, NVP (Viramune®, Viramune XR®)
Rilpivirine, RPV (Edurant®)
Doravirine, DOR (Pifeltro®)

PIs

Atazanavir, ATV (Reyataz®)
Darunavir, DRV (Prezista®)
Fosamprenavir, FPV (Lexiva®)
Indinavir, IDV (Crixivan®)
Nelfinavir, NFV (Viracept®)
Ritonavir, RTV/r**(Norvir®)
Saquinavir, SQV (Invirase®)
Tipranavir, TPV (Aptivus®)

INSTIs

Bictegravir, BIC (Bictarvy®)
Dolutegravir, DTG (Tivicay®)
Elvitegravir, EVG (Vitekta®)
Raltegravir, RAL (Isentress™)

Fusion Inhibitor

Enfuvirtide, ENF/T-20 (Fuzeon®) *

Entry Inhibitor

Maraviroc, MVC (Selzentry®) **
Ibalizumab (Trogarzo®) ***

PK Enhancer

Cobicistat, COBI/c (Tybost®) ****
Ritonavir, RTV/r (Norvir®)

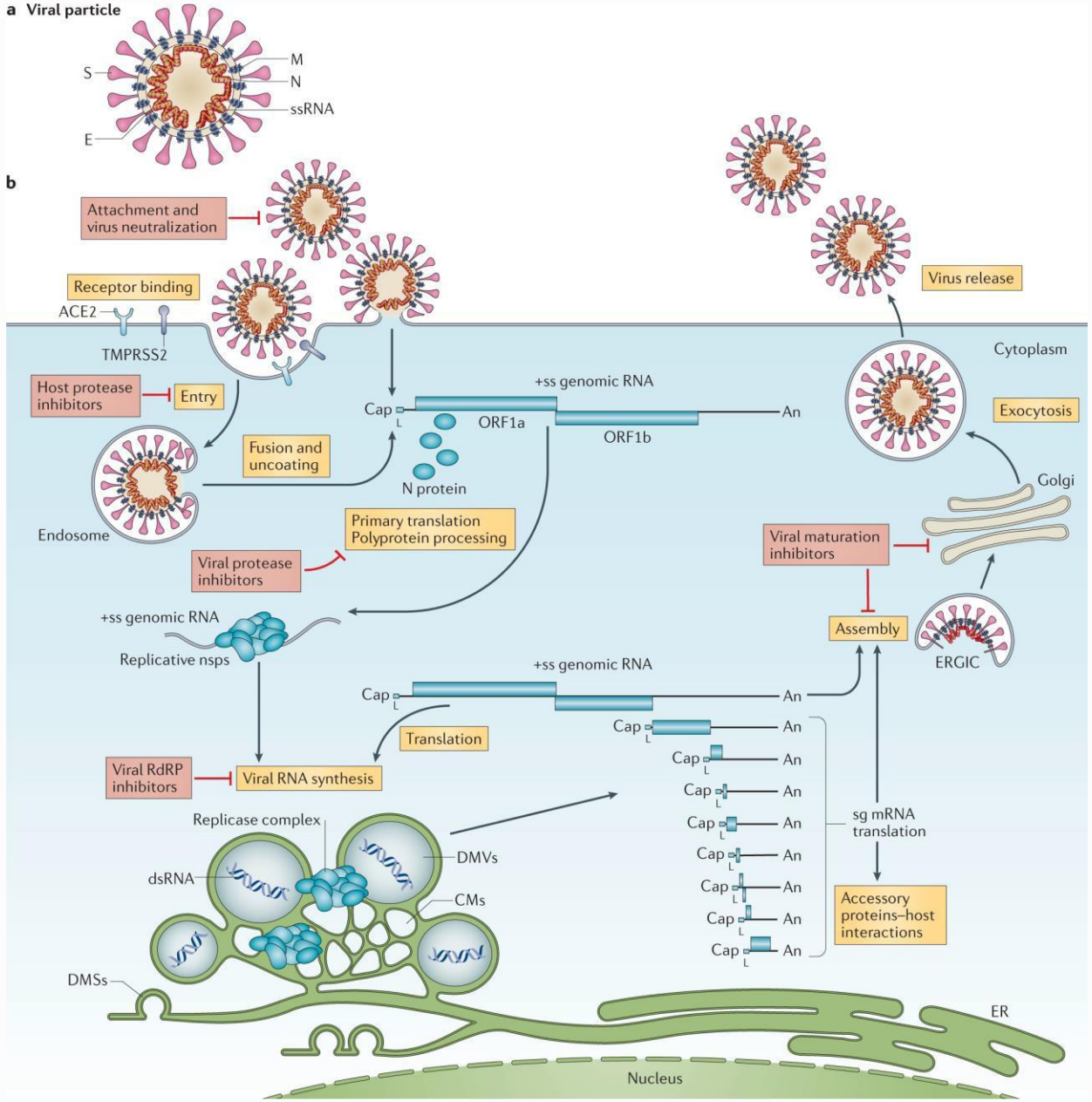
* Binds gp41 (gp41/gp21-CD4)

** CCR5 allosteric modulator

*** anti-CD4 antibody (non-immune suppressive) allosteric modulator

**** Prevents breakdown of Elvitegravir

The coronavirus virion and life cycle



Time to stop using ineffective Covid-19 drugs

Supplementary online Table 1: World Health Organisation recommendations on which medications should or should not be used to treat Covid-19 (adapted from the WHO living guidelines for therapeutics and Covid-19, 14 July 2022)[8]

	Strong recommendation against	Recommendation against	No recommendation for or against	Recommendation for	Strong recommendation for
Non-severe disease	• Convalescent plasma • Hydroxychloroquine • Lopinavir-ritonavir • Colchicine	• Corticosteroids[a] • Ivermectin[b] • Fluvoxamine[b] • Nirmatrelvir and ritonavir[a,c]	• Heparin • Metformin	• Molnupiravir[a,d] • Sotrovimab[a,d,e] • Remdesivir[a,d] • Casirivimab and imdevimab[a,d,f]	• Nirmatrelvir and ritonavir[d]
Severe disease	• Hydroxychloroquine • Lopinavir-ritonavir	• Ruxolitinib and tofacitinib[a] • Ivermectin[b] • Convalescent plasma[b]		• Casirivimab and imdevimab[a,f]	• Corticosteroids • IL-6 receptor blockers or Baricitinib

Note: Severe disease is defined as oxygen saturation <90% on room air, pneumonia or respiratory distress

[a] Conditional recommendation

[b] Only recommended for use within the context of a clinical trial

[c] In non-severe patients at low risk of hospitalization

[d] In non-severe patients at highest risk of hospitalization

[e] Excluding pregnant or breastfeeding women, and children

[f] If seronegative for SARS-CoV-2 antibodies (note: evidence of limited efficacy for omicron BA.1 variant)

[8]Agarwal A, Rochwerg B, Lamontagne F, Siemieniuk RA, Agoritsas T, Askie L, et al. A living WHO guideline on drugs for covid-19. BMJ. 2020;370:m3379.